MIL-C-44313(GL)
30 June 1988

MILITARY SPECIFICATION

COVER, CHEMICAL PROTECTIVE, PATIENT WRAP

This specification is approved for use by the U.S. Army Natick, Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of cover, chemical protective, patient wrap.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-I-1788	-	Ink, Duplicating, Lithographic Process Kit
DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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MIL-B-117 - Bag, Sleeve and Tubing - Interior Packaging
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment, Equipage; General Specification For
MIL-C-44303 - Cloth, Porous, Laminated Fabric System

STANDARDS

FEDERAL

FED-STD-101 - Test Procedures for Packaging Materials
FED-STD-191 - Textile Test Methods
FED-STD-595 - Colors (Requirements for Individual Color Chips)
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY MEDICAL MATERIEL DEVELOPMENT ACTIVIY,
FORT DETRICK, MD 21701-5010

DRAWINGS

60450 FD - Cover, Chemical Protective, Patient Wrap

(Copies of drawing, publications, and other Government documents required by contractors with specific acquisition functions should be obtained from the contracting actvity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case the document shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials. The materials shall be as specified on Drawing 60450 FD and all subsidiary drawings and parts lists pertaining thereto.

3.3.1 Label. Each cover shall have a class 1 identification label conforming to type VI of DDD-L-20, except that the colorfastness requirements shall not apply. The label shall contain the following information with the laminate cloth lot number hand printed or stamped, minimum 10 point characters with Black (No. 37038) ink conforming to type I of TT-I-1788.

COVER, CHEMICAL PROTECTIVE, PATIENT WRAP
CONTRACT NUMBER
NATIONAL STOCK NUMBER
MANUFACTURER
LAMINATE CLOTH LOT NUMBER

3.3.2 Seam sealing tape. The seam sealing tape used to seal the seams connecting the ground sheet to the laminated cloth shall be an ethylene copolymer based hot melt adhesive with vinyl tie coating barrier to high temperature propylene, 1 ± 1/16 inch wide, 0.006 ± 0.0005 inch thick, and a melt temperature (gradient heat bar) of 173° to 175°F nominal.

3.4 Design and construction. The design and construction of the cover shall be as specified herein and as shown on Drawing 60450 FD and all subsidiary drawings and parts lists pertaining thereto.

3.4.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as shown on Drawing 60450 FD. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams occur. Overedge stitching shall be 3/16 to 1/4 inch wide.

3.4.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

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3.4.1.1.1 Repairs of type 301 stitching.

a. When thread breaks skipped stitches, run-offs, or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area, continue over the defective area, and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner.

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.4.1 2 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.4.2 Window sealing. The window shall be heat sealed to the cover in accordance with the applicable drawings, using a heat sealing machine capable of securing the window, in a one step operation, in accordance with the requirements of 3.6. Stitching of the window to the cover prior to heat sealing is prohibited.

3.4.2.1 End tape sealing. The ends of each taped seam shall be independently secured either by a cross-over sealer or low heat hand iron such that there are no lifted ends or corners. In either case, the polyethylene material shall not exhibit any thermal reaction such as puckering, melting, or any sign of degradation.

3.4.3 Repairs to laminate cloth. The following repair method is acceptable provided the contractor performs 100 percent in-process examination of the cover for the repairable defects herein. Repairs shall be made by pinching the material face to face where damaged and stitching 1/16 to 1/8 inch from damaged edge for a length of not more than 1-1/4 inch and tapering each end to form a dart. No more than two repairs shall be allowed on any one cover.

3.5 Carbon tetrachloride absorption. The carbon tetrachloride absorption of the cover shall be not less than 1.8 mg/cm^2 when tested as specified in 4.4.7.1.

3.6 Window heat seal peel strength. The peel strength of the window heat seal shall be no less than 5 pounds per inch of seam width when tested in accordance with 4.4.7.

3.7 Finished measurements. The finished measurements shall be as shown on Drawing 60450 FD.

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3.8 Workmanship. The finished cover wrap shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. The presence of any defect shall be cause for rejection of the first article.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 Pre-production seam sealing system evaluation. Prior to seam sealing any patient wrap end-items, the basic polyethylene material shall be evaluated to assure proper seam sealing machine settings, tape alignment, appearance, and sealability. An individual evaluation shall be conducted per seam sealing machine at the beginning of each day that patient wrap end-items are be to seam sealed. The following constitutes this evaluation:

(1) Seam sealing procedure - Prepare a rectangular specimen of polyethylene material at minimum size 8 x 24 inch (any direction). Single needle sew, using either 301 lock-stitch or 401 chainstitch using size E thread at 8-12 stitches per inch, down the middle of the 8 inch material width for the entire 24 inch length. Operate seam sealing machine according to tape manufacturer's recommended settings and apply middle of seam sealing tape over sewn seam on the dull green side of material. Observe polyethylene material under taped section and along tape sides by through lighting. There shall be no sign of melting, burning, delamination, puckering, or any visible degradation of any kind to the material. Further, the tape shall be crystal clear and be free from any edge puckering throughout the entire length of the specimen.

(2) Tape alignment - At random, fold a section of seam sealed material across the width of tape such that a crease mark is visible when unfolded. Observe that the tape edges remain clear with no sign of delamination at this point. Note that any unclear tape areas signify either delamination or points in which the heat nozzle of the sealing machine are in misalignment with the tape.

(3) Hydrostatic resistance - To determine proper seal, a randomly selected taped section shall withstand a hydrostatic resistance of 50 cm for 3 minutes with no sign of moisture penetration through the seam when tested in accordance with TM 5514. As an alternate, a commercially available hydrostatic machine with a 4-1/2 inch head may be utilized. The taped specimen shall maintain a hydrostatic pressure of 0.75 psi for 3 minutes with no sign of moisture penetration. In either case, the taped material side shall be against the hydrostatic head. In case of dispute at the time of testing, TM 5514 shall be used.

(4) Reporting - If previous taped material observation, tape alignment, and hydrostatic resistance results meet requirements, then record sealing machine settings (temperature, pressure, and speed) and date onto taped material specimen with indelible ink, and continue with daily production utilizing these settings. Repeat all tests and reporting procedure for each day of production. Retain all taped specimens until acceptance of end-items has been finalized.

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4.4.3 In-process inspection. In-process inspection shall be performed on the laminated material during the cutting operations and fabrication of the cover. Any cut, tear, needle chew, hole, burn, or mend shall be cause for rejection of the laminate. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance.

4.4.4 End item critical defect examination. Every end item, prior to the visual and dimensional examinations in 4.4.3 and 4.4.4, shall be examined for the critical defects listed in table I. This 100 percent critical defect examination shall become part of the contractors inspection system or quality program. Any end item found to contain one or more critical defects shall be rejected.

4.4.5 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 10.0 for total (major and minor combined) defects. Any critical defect found during the examination shall be cause for rejection of the lot.

TABLE I. End item visual defects

Examine	Defects	Classification Critical	Major	Minor
Material defects and damages	a. Applicable to laminated cloth.			
	1. Any cut, tear, needle chew, hole, or burn thru one or more layers of cloth:			
	- 1/16 to 1 inch long inclusive		X	
	- more than 1 inch long	X		
	2. Any cut, tear, hole, or burn in knit base fabric:			
	- 1/2 to 1 inch long inclusive			X
	- more than 1 inch long		X	
	3. Any mend:			
	- more than 1/4 inch in longest direction		X	
	- more than two mends per cover		X	
	b. Applicable to cotton/nylon outer shell cloth:			
	1. Any hole, needle chew, cut, tear, or burn.		X	
	2. Mends:			
	- any mend up to 1/4 inch in diameter or length			X
	- more than 1/4 inch in diameter or length		X	
	- more than two mends		X	
	3. Any defect such as a smash, multiple float or slub.			X

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TABLE I. End item visual defects (cont'd)

Examine	Defects	Classification Critical	Major	Minor
Cleanness	a. Spot or stain (grease, oil, ink)			X
	NOTE: Stains attributed to charcoal content of lining should not be scored.			
	b. Thread ends not trimmed or loose thread ends not removed from cover to be scored only when the condition exists on a major portion of the cover.			X
Components and assembly	a. Any component part or required operation omitted, unless otherwise specified.			X
	b. Any assembly or operation not properly performed, unless otherwise classified herein.			X
Cutting	Any component part not cut in accordance with specified requirements.		X	
Seams and stitching	a. Accuracy of seaming:			
	1. Seams twisted, puckered, or pleated (unless otherwise classified herein)			X
	2. Part of cover caught in any unrelated operation or stitching.		X	
	3. Ends of seams or stitching produced with 301 stitch type, when not caught in other seams or stitchings, backtacked less than 1 inch.			X
	4. Stitching overlapping end(s) of thread breaks less than 1 inch.			X
	5. Various shades of thread used on outside of the same cover.			X
	6. Ends of a continuous line of stitching not overlapped or overlapped less than 1 inch.			X

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TABLE I. End item visual defects (cont'd)

Examine	Defects	Classification Critical	Major	Minor
Seams and stitching (cont'd)	b. Gage/margin of stitching:			
	1. Gage of double needle stitching more or less than specified.		X	
	2. Margin of edge or raised stitching more or less than specified:			
	- up to a length of 2 inches inclusive			X
	- beyond 2 inches in length		X	
	c. Open seam:			
	1. Any on joining seam of laminate cloth:			
	- up to a length of 1/4 inch inclusive			X
	- over 1/4 but not greater than 7/8 inch in length inclusive			X
	- more than 7/8 inch in length	X		
	2. Any seam except joining seam of laminate cloth:			
	- up to a length of 1/2 inch inclusive			X
	- more than 1/2 inch in length		X	
	d. Raw edges (outer shell):			
	1. On outside:			
	- up to a length of 1/2 inch inclusive			X
	- more than 1/2 inch in length		X	
	2. On inside:			
	- any raw edge more than 1/2 inch in length (where edge is required to be turned in)			X
	NOTE: Raw edge not securely caught in stitching shall be classified as an open seam.			
	e. Runoffs:			
	1. Joining seams. When resulting in an open seam, use "open seam" classification.			

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TABLE I. End item visual defects (cont'd)

Examine	Defects	Classification Critical	Major	Minor
Seams and stitching (cont'd)	f. Seam and stitch type:			
	1. Wrong seam or stitch type			X
	2. Looper thread of 401 stitch type finishing on outside of cover.		X	
	g. Stitch tension:			
	1. Loose tension, up to 1 inch.			X
	2. Loose tension, more than 1 inch.			X
	3. Tight tension (stitches break when normal strain is applied to the seam or stitching).		X	
	NOTE: Puckering is evidence of tight tension. When puckering is evident, seam shall be tested by exerting pull in lengthwise direction of seam or stitching.			
	h. Stitches per inch (to be scored only when the condition exists on major portion of seam).			
	1. Less than specified:			
	- one or two stitches			X
	- more than two stitches		X	
	2. One or more stitches more than specified.			X
	i. Stitches skipped or broken:			
	- up to a length of 1/2 inch inclusive			X
	- over 1/2, but not greater than 1 inch in length		X	
	- more than 1 inch in length	X		
	j. Heat sealed seams, both window and taped:			
	- any area where the window or tape is not heat sealed	X		
	- any seal twisted or puckered		X	
	- any burn mark or window tape, laminated cloth or ground sheet resulting in a hole	X		
	- resulting in a charred area		X	

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TABLE I. End item visual defects (cont'd)

Examine	Defects	Classification Critical	Major	Minor
Slide fastener	a. Any part of slide fastener assembly omitted, bent, broken, cracked, or otherwise defective, affecting function.	X		
	b. Top ends of tape not turned in and caught in the stitching joining slide fastener to cover.			X
	c. Top stop less than 1/2 inch or more than 5/8 inch from neck edge, or bottom edge of tape more than 1/4 inch from bottom of cover or extending beyond bottom edge.			X
	d. Back edge of slide fastener tape less than 2 or more than 2-1/4 inches from front edge.			X
	e. Shade of tape not as specified.			X
	f. Not functioning properly, failing to effect a secure closure, or to open freely.		X	
	NOTE: The slide fastener shall be fully closed and opened to determine if fastener operates smoothly and provides a secure closure.			
	g. Bottom closed-end bottom stop not attached as specified.		X	
	h. Not specified type or size.		X	
	i. Slider jams or fails to interlock scoops.		X	
	j. Thong omitted.		X	
	k. Fastener tape cut or torn.		X	

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TABLE I. End item visual defects (cont'd)

Examine	Defects	Classification Critical	Major	Minor
Flaps	Hook or pile of fastener tape not as specified.		X	
Labels	a. Labels not positioned and attached to the knit side of lining as specified.			X
	b. Size, identification, or combination label (when applicable) missing, incorrect, or illegible.			X
Drawcords	a. Dipping or impregnation omitted on any end.			X
	b. Missing or not as specified.		X	

4.4.6 End item dimensional examination. The end items shall be examined for the dimensions specified on Drawing 60450 FD. Any dimension not within the specified tolerance shall be scored as a defect. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.7 End item testing. The end items shall be tested for the characteristics listed in table II. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units shall be 2.5.

TABLE II. End item tests

Characteristic	Requirement	Test method	Number determinations per sample unit	Results reported numerically to nearest
Window heat seal peel strength	3.6	5960 1/	4	.1 pound

1/ After the specimen has been under standard conditions for 48 hours the adhesion of the heat sealed seams shall be determined as specified in Method 5960 of FED-STD-191 except the adhesion shall be the lowest individual load

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resistance registered in pounds per square inch of seam width on the autographic chart for 3 inches of seam separation in lieu of the average of 5 highest peak loads of resistance registered for 3 inches of separation.

4.4.7.1 End item carbon tetrachloride absorption testing. Government acceptance testing shall be performed on each lot of the end item for conformance to the requirement of 3.5. Samples shall be forwarded as directed by the contracting activity for testing (see 6.5). The sample unit shall be one completely fabricated cover, and the sample size shall be as follows:

Lot size (covers)	Sample size (No. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Testing shall be in accordance with MIL-C-44303 except that three test specimens shall be selected from each sample unit and shall be from random locations such that each specimen will have been separated from each of the other specimens by at least 1 foot. Any specimen failing to meet the requirement in 3.5 shall be cause for rejection of the lot.

4.4.8 Packaging inspection.

4.4.8.1 In-process testing of heat seal closure seam. Test shall be made daily prior to start of filling and sealing the unit packs to determine that heat sealing equipment is producing seals in accordance with the requirements of 5.1.1. One full width heat seal closure seam shall be made on an empty barrier bag using recommended heat seal settings and tested as specified in 4.5.2. Any test failure shall be cause for readjustment of the sealer settings and retesting until a satisfactory seam is produced.

4.4.8.2 Unit pack examination. Every filled and sealed unit pack shall be examined for the defects listed below. The finding of any defect shall be cause for rejection of the unit pack. Covers from rejected unit packs may be re-unit packed as specified in 5.1.1 and submitted for re-examination.

Defect

Closure seal width not as specified.
Closure seal not continuous.
Tear, cut, or hole in unit pack.

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4.4.8.3 Unit pack leakage testing. Every filled and sealed unit pack, prior to packing as specified in 5.2, shall be tested as specified in 4.5.1 for conformance to the leakage requirements in 5.1.1. Any unit pack which fails the test in 4.5.1 shall be opened and the cover shall be removed, thoroughly dried, re-unit packed as specified in 5.1.1, and retested as above.

4.4.8.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and unit pack)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Content	Number per container is more or less than required.

4.4.8.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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4.5 Methods of inspection.

4.5.1 Leakage test. The unit pack shall be tested for leaks in accordance with the hot water technique of Method 5009 of FED-STD-101. All samples shall be conditioned at ambient conditions for at least 4 hours prior to performing the hot water test. The time that the unit pack and all processing materials have been maintained at ambient conditions prior to or during the processing period may be considered a part of the conditioning time. Bubbles which may appear on the surface of the unit pack but are not released at a slowly decreasing rate are not to be construed as an indication of failure. A steady stream or recurring succession of bubbles from any surface or seam shall be considered a failure.

4.5.2 Sealed seam test. The test shall be performed in accordance with Method 2024 of FED-STD-101 at normal room temperature with a static load of 50 ounces using fabricated bag specimen procedure. A 5-percent reduction in static load is permitted when room temperature in the test area exceeds 90°F. Any seal separation shall be considered a test failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. One cover and face plate shall be unit packed in a heat sealable, flexible, watervaporproof, greaseproof, barrier bag conforming to type I, class E, style 1 of MIL-B-117. The cover shall be folded so that the camouflage surface is exposed. The dimensions of the folded cover shall be approximately 16 inches in length, 15-1/2 inches in width, and 5 inches in depth. The face plate shall be folded to the smallest dimensions and placed between the folds of the cover. The barrier bag shall be flat style, having inside dimensions of approximately 19 inches by 18 inches. The outer surface of the barrier bag shall be uniformly colored overall with a color in the range of 34079 to 34087, inclusive, of FED-STD-595. The barrier bag shall be sealed at a vacuum level of 17 inches of mercury (± 2 inches). All four sides shall be heat sealed. The seals shall be 1/2 ± 1/8 inch width when fabricated by jaw-type sealers, and 3/16 ± 1/16 inch width when fabricated by dielectric, impulse or ultrasonic processes. Seals shall show no separation when tested in accordance with 4.5.2. The filled and sealed unit pack shall not leak when tested in accordance with 4.5.1.

5.2 Packing. Packing shall be level A or B, as specified (see 6.2).

5.2.1 Level A packing. Six covers, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be filled with a box liner, conforming to type CF, class weather-resistant, variety DW, grade 275 of PPP-B-636. Corrugated fiberboard pads shall be placed on the bottom and top of

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the box. Inside dimensions of the box shall be 19-3/4 inches by 15-3/4 inches by 14-1/2 (± 1/8) inches. Each shipping container shall be closed in accordance with method III and waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.8.4. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Six covers, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to RSC-L, grade V3c of PPP-B-636. The inside of each shipping container shall be filled with a box liner, conforming to type CF, class weather-resistant, variety DW, grade 275 of PPP-B-636. Corrugated fiberboard pads shall be placed on the bottom and top of the box. Inside dimensions of the box shall be 19-3/4 inches by 15-3/4 inches by 14-1/2 (± 1/8) inches. Each shipping container shall be closed in accordance with method III and waterproofed in accordance with method V as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.8.4.

5.3 Palletization. When specified (see 6.2), covers, packed as specified in 5.2.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia, of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding O or P. Pallet pattern shall be number 95 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The cover is intended for use by military personnel to provide non-ambulatory patients protection against chemical agents.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.2, 4.3, and 6.4).
c. Selection of applicable levels of packing (see 5.2).
d. Type and class of unit load required (See 5.2.1).
e. When palletization is required (see 5.3).

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6.3 Sample and shade samples. For access to sample of the end item, and shade samples address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Address. Samples for testing should be addressed to the contracting activity issuing the invitation for bids.

6.6 Subject term (key word) listing.

Chemical protective
Cover
Patient
Wrap

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

Project No. 8415-0618

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44313(GL)	Cover, Chemical Protective, Patient Wrap

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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